Sahaj Bhadja

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Summary

Software Engineer with 2+ years of experience building **distributed data systems**, **backend services**, and responsive web platforms using Java, Python, TypeScript, SQL, and cloud infrastructure.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Reduced dashboard latency from 1.2s to 350ms through **SQL indexing and query optimization**, designing normalized schemas and row-level isolation for a multi-tenant education platform supporting 500+ users reliably.
- Achieved 99%+ production reliability across 5K+ monthly jobs through **incremental data pipelines**, validation, reconciliation, error handling, cloud storage, CI/CD releases, and stable production observability reliably at scale.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Reduced feedback turnaround by 60% through a **production Python platform** supporting 150+ instructors and 10K+ daily requests with REST APIs, AWS RDS, secure ingestion, and CI/CD workflows.
- Processed 1K+ files per batch using **Python multithreading**, validating heterogeneous inputs while isolating failures and preserving responsive customer workflows during concurrent model execution and production traffic reliably.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Improved API response times by 30% through **maintainable microservices**, indexing, caching, pagination, and filtered payloads while troubleshooting integrations across Windows, Linux, and macOS production environments reliably at scale.
- Automated builds and 20+ unit/integration suites through **Jenkins and GitHub**, adding failure reports and cross-platform validation across Linux, Windows, and macOS with repeatable nightly workflows reliably daily.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Synchronized 2M+ records through an eight-service **microservice platform** covering authentication, resources, customers, billing, leads, permissions, and notifications with Node.js, WebSockets, and Azure SQL across locations and workflows.
- Reduced resource wastage by 42% through **real-time data synchronization**, implementing role-based access, WebSocket updates, automated tests, Azure monitoring, and Docker/Jenkins deployments across eight modular product service domains.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Delivered **sub-200ms Flask APIs** across 50K+ booking records, implementing pricing, coupon validation, payment webhooks, and Firebase status synchronization while using transactions and locking to prevent conflicting reservations.
- Reduced manual refund handling by approximately 70% through **payment workflow automation**, supporting refunds, disputes, and consistent status while presenting architecture and backend integration workflows directly to clients reliably.

Skills

Languages: Java, TypeScript, JavaScript, Python, SQL, C++, Bash
Backend & Frontend: Node.js, React, FastAPI, Flask, REST APIs, WebSockets, Microservices, API Design
Data & Systems: PostgreSQL, MySQL, MongoDB, Redis, Azure SQL, Data Modeling, Distributed Systems, Caching, Pagination
Cloud & Delivery: AWS, AWS RDS, AWS S3, AWS Glue, Azure, Docker, Kubernetes, CI/CD
Engineering Quality: Data Structures, Algorithms, Git, GitHub, Unit Testing, Integration Testing, PyTest, Mockito, Code Review, Data Validation

Projects

Unified Stakeholder Profile Store - Design Concept

- Designed a **distributed profile-store concept** supporting identity resolution, versioned attributes, transactional events, deduplication rules, role-based access, audit history, dynamic segments, and low-latency customer-profile retrieval reliably at scale.

Real-Time Profile Serving Layer - Design Concept

- Designed a **real-time serving layer concept** using event-driven ingestion, partitioned consumers, idempotent updates, cache invalidation, searchable profile projections, freshness metrics, replayable events, and failure-isolated processing workflows reliably.

Chat2SQL | Live

- Unified **10+ database systems** in a Java and TypeScript workspace extending Chat2DB, integrating schema exploration, SQL generation, structure synchronization, and Docker-based local deployment across Windows, macOS, and Linux environments.